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REVIEW **Open Access**

Yeast-based nanofibers technologies for sustainable fruit fly pest management: a review

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Abstract
Numerous species within the family *Tephritidae* are major agricultural pests and infest a broad array of fruits and vegetables in tropical and subtropical regions. As a result, they pose a serious threat to global fruit production. Traditional pest management techniques often face limitations due to their harmful effects on the environment and reduced efficacy in variable field conditions such as rain, humidity, temperature, and light. Recently, electrospinning technology has demonstrated remarkable potential in the preparation of sustained carriers. While several review articles have addressed the use of electrospun nanofibers for pest control, no comprehensive review has specifically focused on yeast-based nanofibers in fruit fly management. This review highlights the significance of yeast-based nanofibers produced via electrospinning as a promising eco-friendly approach for fruit fly pest management. The unique properties of electrospun nanofibers such as large surface area, high porosity, and controlled release capabilities, improve the stability, longevity, and delivery efficiency of yeast-derived attractants. This novel approach addresses the issue of rapid volatilization and environmental breakdown of volatile compounds, prolonging the shelf life and increasing the field stability of biological attractants. This review synthesizes current knowledge yeast metabolism, genetic engineering of yeast for pest control, and the advancements in electrospinning for controlled release. The technical considerations are critically discussed, including polymer and solvent selection, and the impact of electrospinning on yeast viability and morphology. While emphasizing the multidisciplinary promise of this approach, we also identify key research gaps and future directions, such as optimizing encapsulation protocols, validating field performance, and addressing potential ecotoxicological impacts, to accelerate the translation of yeast-based nanofiber technology into practical, scalable, and sustainable solutions for fruit fly control.

Keywords Electrospinning, Pest management, Sustainable, Encapsulation, Biopolymer, Nanofibers

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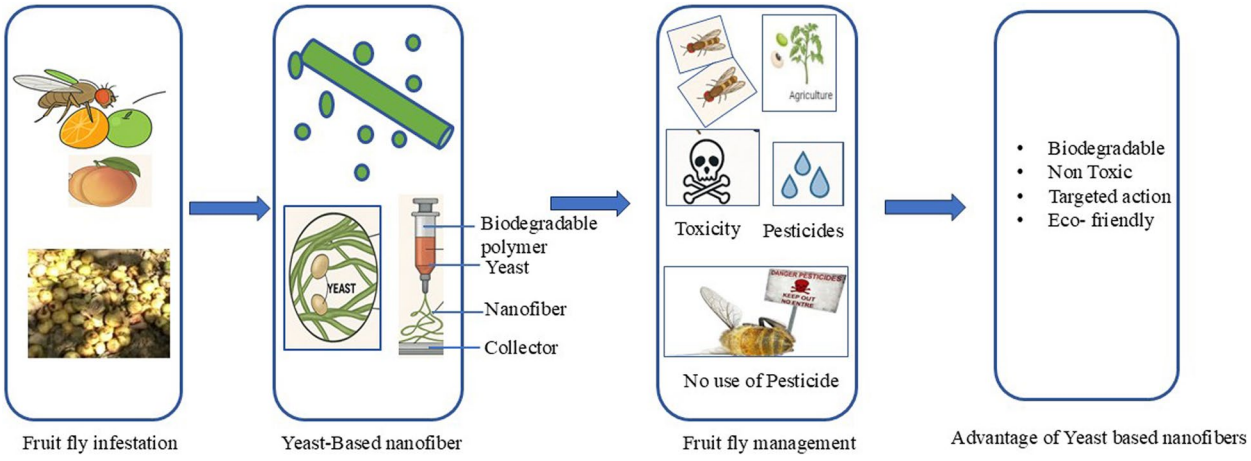
Graphical abstract



Introduction

Fruit flies (Diptera: Tephritidae) represent one of the most economically significant groups of insect pests globally, posing severe threats to fruit crop production across diverse geographical regions [1, 2]. Key destructive species include the Mediterranean fruit fly (*Ceratitis capitata*), Oriental fruit fly (*Bactrocera dorsalis*), Mexican fruit fly (*Anastrepha ludens*), and Olive fruit fly (*Bactrocera oleae*). The economic consequences of these pests go beyond direct crop damage, encompassing stringent quarantine restrictions on infested areas, necessitating costly pre-export treatments, and serving as reservoirs for pest spread [3]. Among the more than 5,000 described fruit fly species, approximately 200 are of major economic importance because of their impact on crop yields and their ability to damage fruits and vegetables [4]. Furthermore, over 250 species are classified as quarantine pests in regions such as the European Union, highlighting their global significance in agriculture, international trade, and regulatory frameworks [4, 5].

Effective and environmentally benign management of tephritid fruit fly populations is a persistent challenge. Current strategies include pest-resistant crops, field sanitation, and the sterile insect technique (SIT) [6]. A comparative analysis of these management strategies, including their advantages and limitations, is presented in Table 1. Attractant-based traps, such as those utilizing methyl eugenol and cue lure, effectively target male flies, reducing mating opportunities and population growth [10]. Broader-spectrum food attractants have also been developed to target both male and female flies [11]. Among natural attractants, the intricate interactions between yeasts and tephritid fruit flies, while less extensively studied than those with drosophilids [12], offer significant potential. Crucially, yeasts like *Saccharomyces* and *Candida* produce volatile organic compounds (VOCs) during fermentation that mimic natural cues associated with food sources and oviposition sites, thereby attracting fruit flies [13]. Harnessing these VOCs for 'attract-and-kill' lures and improving mass-rearing techniques for SIT programs are active areas of research.

A significant limitation in current attractant-based pest management is the instability and rapid volatilization of active compounds under variable environmental conditions (e.g., rain, humidity, temperature, light). This necessitates frequent reapplication, increasing costs and labor, and limiting sustained control. Nanofiber encapsulation method offers a compelling solution to this challenge. While the literature on electrospun fibers for diverse applications, including controlled release of pesticides and repellents, is growing, there remains a notable research gap concerning the specific application of yeast-based electrospun nanofibers for fruit fly management with control release of attractant.

The focus of this review is to provide an overview of the use of yeast-based nanofibers produced via electrospinning for the sustainable management of fruit fly pests. The aim of this review is to examine the integration of yeast attractants produced through electrospinning to enhance efficiency, persistence, and controlled release of attractants thereby addressing the critical need for eco-friendly and long-lasting pest control solutions.

Electrospinning for controlled release in pest management

The development of suitable sustained-release carriers is crucial for the success of pesticide encapsulation and enhancing the efficacy and longevity of pest control agents [14]. Among the various technologies explored, electrospinning has emerged as one of the most versatile and promising methods for producing sustained-release carriers, particularly through the incorporation of

Table 1 Comparative analysis of tephritid management strategies

Strategy	Mechanism of Action	Advantages	Limitations	Environmental Impact	References
Sterile Insect Technique (SIT)	Mating with sterile males prevents the production of viable offspring.	Species-specific, environmentally friendly, and integrates well with other controls.	High cost, logistically complex, and requires high release ratios to compensate for reduced male competitiveness.	Minimizes insecticide use and has no negative impacts on non-target species.	[7]
Pest-Resistant Crops	Plant traits deter pests (antixenosis) or directly harm them (antibiosis), reducing infestation.	Reduces the need for external controls, can be highly effective.	Longer time for developing new plant varieties, public perception and regulatory barriers for GE crops.	Reduces the use of chemical pesticides but raises concerns about gene flow and non-target effects.	[8]
Field Sanitation	Physical removal or destruction of infested fruit, interrupting the pest's life cycle.	Highly accessible, non-chemical, and a fundamental component of all IPM programs.	Extremely labor-intensive, ineffective in areas with unmanaged wild hosts.	No negative environmental impact; promotes soil health and nutrient recycling.	[9]

biodegradable materials, thereby paving the way for eco-friendly pest management solutions [15].

Electrospinning is a fiber fabrication technique that uses electrostatic forces to produce continuous and tiny fibers at both the nano and microscales [15]. By adjusting process parameters—such as voltage, flow rate, solvent type, and collector design—nanofibers can be precisely engineered to meet specific material properties and application requirements [16, 17]. Several electrospinning variants, such as solution electrospinning (directly mixing active compounds with polymer solutions), melt electrospinning (uses a polymer melt instead of a polymer solution), coaxial electrospinning (producing core–shell nanofibers for precise release control), and emulsion electrospinning (enabling simultaneous encapsulation of hydrophilic and lipophilic agents), have been utilized to enhance loading efficiency and controlled release behavior [18]. The schematic illustrations of various electrospinning methods have been presented in Fig.1. The nanofibers generated via electrospinning possess unique properties, including high surface area-to-volume ratio, porosity, and mechanical flexibility [19]. These characteristics make them ideal candidates for encapsulating bioactive substances and enhancing their stability, bioavailability, and controlled release. The small pore sizes and the ability to form dense matrices make electrospun nanofibers a highly effective material for encapsulation [20]. While several encapsulation technologies, such as metal-organic frameworks (MOFs), nanoparticles, and microcapsules, have been used for sustained pheromone release [21], due to restricted loading capacity, sub-optimal release kinetics, complex preparation methods, and high production costs, their use often remains limited. For example, the loading capacity of pesticides and other agrochemicals is highly dependent on the type of MOF utilized. Reis et al. [22] reported a loading capacity of 45% for UiO-66, whereas an iron-based MOF demonstrated only 16.2% [23]. A major drawback of MOFs is the initial burst release, where a substantial proportion of the active ingredient is rapidly discharged following application, potentially resulting in environmental contamination and phytotoxicity [22]. Moreover, conventional MOF synthesis often requires toxic organic solvents such as dimethylformamide (DMF), N,N-diethylformamide (DEF), and dimethyl sulfoxide (DMSO), which are cytotoxic and can compromise or destroy yeast cells during synthesis [24]. In addition, the high cost of raw materials, particularly organic linkers and certain metal precursors—renders large-scale MOF production economically challenging, posing a significant barrier to their widespread adoption in agriculture [25]. In contrast, electrospun nanofibers offers a cost-effective, scalable, and flexible alternative for developing eco-friendly carriers [26].

In recent years, significant advancements have been made in the use of electrospun nanofibers as carriers for the incorporation and release of pheromones for plant protection [27]. For instance, Czarnobai De Jorge et al. [28] used electrospinning to produce nanoscale dispensers for attract-and-kill strategies. The study demonstrated an impressive efficacy of >87% mortality in 84-day tarsal-contact bioassays against target insects. Similarly, Xiang et al. [27] designed a biodegradable controlled-release system that combined the pheromone of the Oriental fruit moth (*Grapholita molesta*) with codlemone using cellulose acetate nanofibers. Field trials demonstrated that the fibers remained attractive for approximately six weeks, highlighting their practical applicability. For Olive fruit fly (*Bactrocera oleae*) management, Kikionis et al. [29] fabricated fiber mats from inexpensive, biodegradable polyhydroxybutyrate. These carriers effectively attracted male fruit flies for more than 16 weeks, outperforming commercial positive controls and underscoring the long-term efficacy of this approach. Electrospinning has also been employed against Grapevine moth (*Lobesia botrana*), where nanofiber-based dispensers achieved a mating disruption effect comparable to that of commercial formulations for at least four weeks [30].

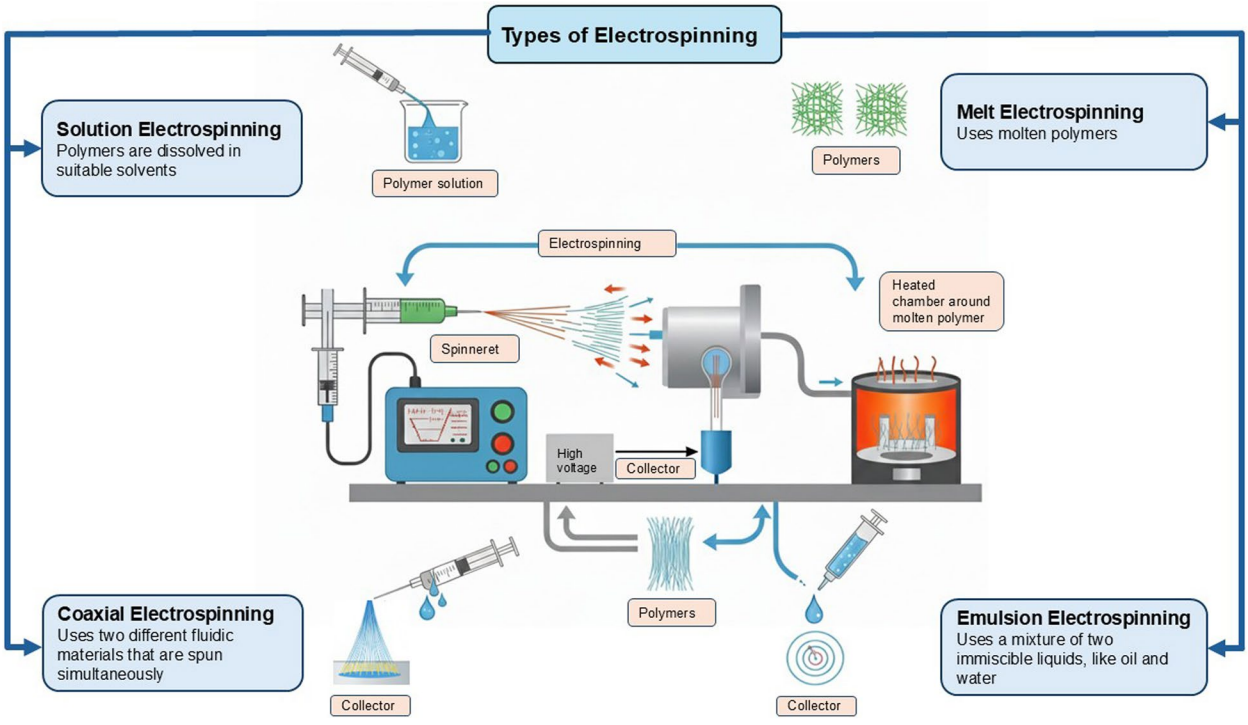


Fig. 1 Schematic representation of different electrospinning techniquesSchematic representation of different electrospinning techniques

Beyond pheromones, electrospinning has been utilized to encapsulate volatile essential oils, such as peppermint (*Mentha piperita*) and sage (*Salvia officinalis*), into nanofibers. In studies by Allahvaisi et al. [31, 32], these oil-loaded nanofibers exhibited sustained fumigant activity against pests such as the Indian meal moth (*Plodia interpunctella*), maintaining insecticidal efficacy for several weeks—substantially longer than the activity of unencapsulated oils. These findings demonstrate the versatility and promise of electrospinning as a sustainable platform for pest management. Recent development in the use of polymer-based delivery system for pest control is summarized in Table 2, which highlights the types of polymers, encapsulation techniques, and their benefits in enhancing stability, targeted release, and efficacy of bioactive agents.

Yeast metabolism and volatile attractants

Yeasts, primarily heterotrophic organisms belonging to the Ascomycota and Basidiomycota phyla within the fungi kingdom, are unicellular microorganisms typically ranging from 3 to 10 µm in length and 2–7 µm in width. They reproduce sexually (via spore formation) or asexually (via budding, e.g., *Candida*, *Saccharomyces*, *Pichia*, *Rhodotorula*, or fission, e.g., *Schizosaccharomyces*) [41]. The ecological relationship between yeasts and fruit flies is well-established, particularly with *Drosophila* species. Yeasts constitute a critical dietary component for *Drosophila* larvae, essential for their development [42]. Furthermore, *D. melanogaster* olfactory receptors have been shown to specifically respond to yeast-derived volatiles [43]. This fundamental biological requirement and olfactory sensitivity make yeasts, and their metabolic byproducts, highly attractive candidates for development into bait-based control strategies targeting fruit flies.

During primary and secondary metabolism, microorganisms, including fungi, bacteria, and particularly yeasts, produce a diverse array of VOCs [44]. Figure 2 illustrates yeast metabolism and the associated production of VOCs. The main function of the VOCs arises from the interactions between microorganisms, generally between bacteria and fungi through cross-communication [45]. Many bacterial species produce VOCs with strong antifungal activity. For example, VOCs emitted by *Pseudomonas palleroniana*, several *Bacillus* strains, and *Paenibacillus* sp. B3a have been shown to inhibit the growth of the soil-borne fungus *Rhizoctonia solani*, a major pathogen of potato [46]. Similarly, fungi also exploit VOCs to antagonize bacteria. The genus *Trichoderma*, for instance, produces a diverse range of volatiles, including C8 alcohols and terpenoids, which have demonstrated potent antibacterial properties [47]. Beyond direct antimicrobial effects, VOCs play an essential role in priming neighboring microorganisms against potential threats. A well-documented example is the volatile blend released by *Bacillus subtilis* GB03, which protected *Arabidopsis* seedlings against *Botrytis cinerea* infection [48]. Such VOC-mediated signaling acts as a chemical

Table 2 Advanced polymer-based systems for pest management

Encapsulation Method	Polymer/Material Used	Target Pest	Outcome/Key Findings	Applications	Reference
Emulsification	PLGA + Neem oil	House dust mites (*Dermatophagoides pteronyssinus*)	Enhanced insecticidal activity with controlled release of neem oil	Miticidal purposes	[33]
Electrospinning	PCL and PEA + Neem oil	Yellow fever mosquito (*Aedes aegypti*)	Thermally stable nanofiber mats with favorable physical and chemical properties	Mosquito control	[34]
Electrospinning	PLA + Columbia Blue dye	Whiteflies	High solubility of dye in fibers; increased pesticide dosage; improved mortality rate	Insecticidal delivery system	[35]
Electrospinning	Polyvinyl polymer + Insect sex pheromone	Codling moth *(Cydia pomonella* (L.))	Enhanced deposition and retention of sex pheromone microcapsules on plant leaves	Lepidopteran pest management	[36]
Electrospinning	PCL, CA, PLA + OFM pheromone + Codlemone	Oriental fruit moths (*Grapholita molesta*)	Controlled pheromone release; 10 mg CA-SP most attractive; 6-week effective duration	Sex pheromone-based plant protection	[27]
Electrospinning	PCL + 1,7-dioxas-piro[5.5]undecane + (Z)−7-Tetradecenal	Olive fruit fly *(Bactrocera oleae)*, Olive moth (*Prays oleae)*	Slow-release biodegradable system; effective attraction of male pests	Fruit fly field trapping	[29]
Electrospinning	Nylon-6,6 + Picaridin	—	Repellent nanofibers with unchanged fiber morphology; long-term release under ambient conditions	Insect repellent	[37]
Electrospinning	PLLA + IR3535	Mosquitoes	IR3535 promotes PLLA crystallization; forms long-lasting orthorhombic α-crystals	Mosquito repellent	[38]
Electrospinning	PET microfibers + DEET + Picaridin	Mosquitoes	Long-lasting repellency achieved	Mosquito repellent	[39]
Electrospinning	PAAm/Yeast nanofiber + F127-DM-PEO nanofibers	*Pseudomonas* spp. + Mosquito larvae	Sustained release of mosquito larvicides over extended period; mosquito-borne disease control	Mosquito control	[40]

PLGA Poly(lactic-co-glycolic acid), *PCL* Poly(caprolactone), *PEA* Pea protein, *PLA* Polylactic acid, *CA* Cellulose acetate, *PLLA* Poly(L-lactic acid), *PET* Polyethylene terephthalate, *DEET* N, N-diethyl-m-toluamide, *PAAm* Polyacrylamide, *DM-PEO* Dextran-modified poly(ethylene oxide)

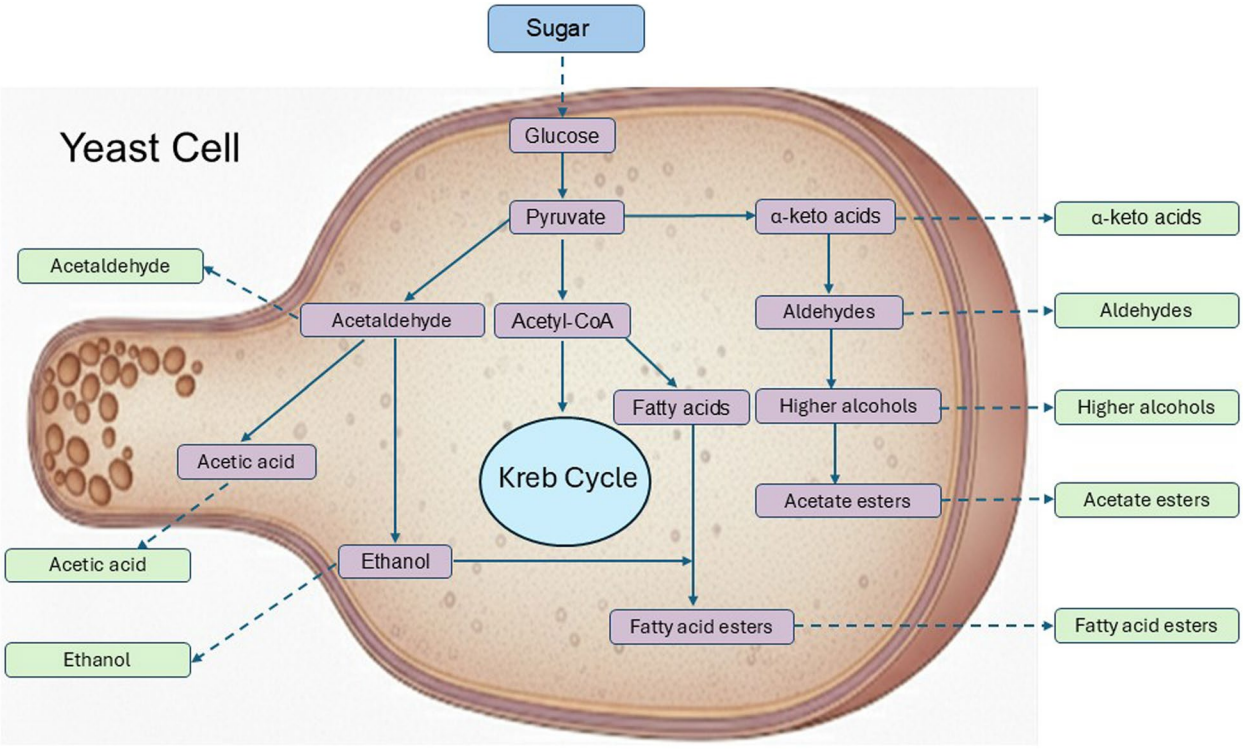


Fig. 2 Overview of yeast cell metabolism for the production of volatile organic compounds. Solidlines denote biochemical pathways (not to scale with the number of individual reactions),while dotted lines indicate the import or export of metabolites

warning system, enabling microorganisms to coordinate community-wide defense responses efficiently.

The VOCs generated by specific strains of *Saccharomyces cerevisiae* have shown strong attractant activity for fruit flies. Such fly attracting yeast VOCs include 2-phenethyl alcohol (2-PEA), ethanol, and acetic acid [49]. Furthermore, studies have shown that yeast-derived volatiles can enhance mating success by increasing female sexual receptivity [50, 51]. Yeast VOCs are usually non-toxic to higher organisms making them biologically safe pest control agents [52]. However, a major challenge of yeast produced volatiles is sustaining their attractiveness, as it changes significantly with yeast's growth phase and environmental conditions [53]. This limitation highlights the importance of developing advanced delivery systems that are capable of controlled and sustained release of attractant compounds and maintaining viability of yeast cells under variable environmental conditions.

Genetic engineering of yeast for enhanced pest control

Genetically engineered yeasts are emerging as a promising tool to produce specific compounds or deliver biological agents for insect pest control, offering multiple avenues beyond their inherent attractant properties. For instance, when genetically modified *S. cerevisiae* express double-stranded RNA (dsRNA) targeting the *yTub23C* gene in *Drosophila suzukii*, led to impaired locomotion, decreased survival, and reproductive fitness in the fruit fly [54]. Similarly, mosquito larvae and adults reared on diets containing yeast strains expressing dsRNA exhibited severe neurological impairments and decreased expression of target genes, demonstrating the potential for yeast-mediated delivery systems for larvicides [55–59]. Yeast-based RNA interference (RNAi) delivery offers several advantages, including oral administration, protection of dsRNA from environmental degradation, and scalability through fermentation. Recent studies have demonstrated that yeast formulations expressing species-specific dsRNA can significantly increase mortality in *D. suzukii* [60], with ongoing efforts to extend this approach to other pests. However, the efficiency of RNAi-based yeast insecticides varies across species due to factors such as gut nucleases, microbiome interactions, and environmental nucleolytic activity, emphasizing the need for careful target selection and optimized formulations [61, 62].

Yeasts have also been engineered to disrupt insect reproduction by diverting males towards artificial sex pheromones. This 'attract-and-disrupt' approach confuse male insects and thus interfere with successful mating. For example, *S. cerevisiae* engineered with sex pheromone biosynthesis genes (*Gm-DES* and *Gm-FAR*) from the Oriental fruit moth (*Grapholita molesta*) successfully replicated the moth's pheromone production pathway, offering an eco-friendly approach to insect pest control [63]. Likewise, genetically engineered *Yarrowia lipolytica* producing pheromones has proven to be an efficient and species-specific strategy for trapping the cotton bollworm (*Helicoverpa armigera*) [64]. Moreover, yeast-derived pheromones have demonstrated performance equivalent to synthetic pheromones in both monitoring and disrupting the mating of *H. armigera* [65]. Furthermore, yeasts such as *Metschnikowia pulcherrima* and *Cryptococcus tephrensis*, genetically engineered with codling moth granulovirus (CpGV), have been shown to reduce the survival of apple tree pest larvae (*Cydia pomonella*), demonstrating their potential as carrier for entomopathogenic agents [66].

Yeast-based protein delivery has also been investigated as a strategy for pest control. For instance, the expression of insecticidal proteins from *Bacillus thuringiensis* (Bt) in *S. cerevisiae* demonstrated that yeast could function as bioinsecticides [67]. While protein-based strategies can be effective, concerns regarding non-target effects and regulatory challenges must be addressed before large-scale implementation. Another promising avenue involves engineering yeasts to enhance VOC production for behavioral manipulation of insect pests. Naturally associated yeasts often elicit stronger insect attraction than *S. cerevisiae* [68], with studies confirming their role in attracting *Vespula pensylvanica* [69], scolytine bark beetles [70], and *D. suzukii* [53]. Engineering yeasts to overproduce specific VOCs or tailored blends could improve lure specificity and shelf-life compared to wild strains. Recent Studies on the use of genetic engineered yeast approaches for pest control are provided in Table 3.

These advancements in yeast genetic engineering underscore their versatility as biological tools. However, the effective deployment of such engineered yeasts in the field necessitates robust delivery systems that can maintain cell viability, control release kinetics, and protect the engineered strains from environmental stressors. This is where nanofiber encapsulation via electrospinning becomes particularly relevant, providing a protective and controlled environment for these advanced biological agents.

Yeast encapsulation for stability, delivery, and pest management

The inherent instability and rapid degradation of yeast-based attractants in ambient conditions pose significant limitations for their widespread field application in pest management. Electrospinning- a process that produces ultrathin fibers (diameter in the submicron range) using a high voltage external electric field- has gained substantial interest in encapsulation of diverse bioactive compounds and delivery system production [73]. Encapsulation of yeast within electrospun nanofibers provides several

Table 3 Applications of genetic engineering in yeast for pest management

Yeast species	Mode of genetic engineering	Target pest(s)	Outcome	Reference
Saccharomyces cerevisiae	Genetically modified yeast express dsRNA targeting y-Tubulin	*Drosophila suzukii*	Decreases larval survivorship and reduces locomotor activity and reproductive fitness in adults of *D.suzukii*	[54]
S. cerevisiae	Yeast engineered to produced short hairpin RNAs (shRNAs)	*Aedes aegypti (Mosquito)*	Silenced target gene expression, disrupted neural development, and highly significant larval mortality	[55]
S. cerevisiae	*S. cerevisiae* engineered to produce short hairpin RNAs (shRNAs) for silencing of actin (Sac1), leukocyte receptor complex member (lrc), and offtrack (otk) genes	*Anopheles gambiae*	Formation of user-friendly dried inactivated yeast interfering RNA tablet formulations, 100% larval mortality in laboratory trials, biorational tool for mosquito.	[57]
S. cerevisiae	Express dsRNA (RNAi) in yeast; formulate as attractive sugar bait	*Drosophila suzukii*	Targeted gene knockdown, increase fly mortality (96%), in lab trails. Species specific sustainable SWD control	[59]
Yarrowia lipolytica, *S. cerevisiae*	Metabolic engineering to express moth's pheromone components	*Helicoverpa armigera*, *Spodoptera frugiperda*,	Yeast-derived pheromones attract moth males and disrupt mating in field and equally effective synthetic pheromones cost effective and scalable fermentation	[64]
S. cerevisiae	Heterologous expression of *Bacillus thuringiensis* (Bt) insecticidal proteins CryIA(a) protein	*Plutella xylostella*	First time Functional expression of large Bt proteins (CryIA(a) in yeast, potential for production yeast based biopesticides	[67]
Wickerhamomyces rabaulensis, *Hanseniaspora guilliermondii*	Overproduction/tuning of volatile organic compounds (VOCs)	*Carpophilus truncatus*	Improved lures/monitoring; potential for mass-trapping strategies using blend of yeast VOCs.	[71]
Kluyveromyces lactis	Genetically engineered yeast strains that express synthetic protozoacidal lytic peptides	*Coptotermes formosanus Shiraki*	Destruction of the gut tissue of Termite. Can be applied for termite control	[72]

advantages that enhance performance and expand their applications.

The nanofiber matrix functions as a protective barrier that shields yeast cells from diverse environmental stressors (e.g., desiccation, UV radiation, extreme temperatures, pH fluctuations), thereby enhancing encapsulation efficiency and cell survival. Under severe desiccation, for instance, proteins denature, cellular membranes collapse, and metabolic activity ceases [74]. Similarly, UV radiation induces DNA damage through pyrimidine dimer formation, which disrupts replication and transcription and can result in mutations or cell death [75]. Elevated temperatures denature essential proteins and enzymes, destabilize membrane fluidity, and ultimately lead to cell lysis [76]. Yeast cells also require a regulated intracellular pH (6.8–7.2) for optimal function; extreme external pH shifts can denature enzymes, impair nutrient transport, damage membranes, and compromise growth and viability [76].

Encapsulation within a nanofiber matrix mitigates these stress-induced damages through multiple protective mechanisms. The dense, porous, three-dimensional network physically immobilizes the cells and limits their direct exposure to external insults. This barrier is particularly effective against UV radiation, which can be partially absorbed or scattered by the matrix [77]. The microenvironment created by the nanofibers also reduces desiccation by slowing moisture loss, while the polymer composition can buffer against sudden pH shifts, thereby minimizing metabolic stress [78]. In addition, immobilization prevents cell aggregation, ensuring uniform spatial distribution that supports consistent activity in applications such as fermentation and bioremediation [79, 80]. Despite serving as a physical shield, the high porosity and large surface area of nanofiber mats allow efficient exchange of nutrients (e.g., sugars, oxygen) and removal of metabolic byproducts. This balance between protection and permeability maintains metabolic activity and enhances long-term cell viability, even under otherwise deleterious environmental conditions [77].

In the context of fruit fly management, the encapsulation of yeast within nanofiber matrices presents several promising opportunities. Yeast cells naturally produce a variety of VOCs that function as semiochemicals (chemical substances or mixture that transmit signals between organisms) to attract fruit flies [81–83]. Encapsulation offers effective protection of these yeast cells, ensuring their continued viability and growth kinetics even under harsh environmental conditions [77, 84]. This controlled delivery system not only enables prolonged and consistent release of attractants but also enhances their duration and effectiveness [14]. Although encapsulation mechanism is mainly used for attractants, it can also be tailored for genetically modified yeasts for synthesis of repellent compounds, or combined with synthetic repellents, enabling dual-purpose pest control strategies. Several studies have shown that encapsulated yeast, specifically *S. cerevisiae*, can serve as a novel and effective

delivery vehicle for entomopathogenic fungi or bacteria (e.g., *Beauveria bassiana*) to fruit fly pests [85–87]. This 'attract-and-infect' approach exploits the natural feeding behavior of fruit flies. In 'attract-and-kill' strategy of pest control, yeast-nanofiber formulations can be combined with insecticides, where the attractant lures the pest to insecticides encapsulated within a matrix, thereby minimizing broad-spectrum use of which can harm beneficial insects and environment. The core principle of yeast encapsulation via electrospinning involves dispersing yeast cells within a polymer solution, which is then subjected to an electric field to form continuous nanofibers (Fig. 3).

Yeast-based nanofibers represent a promising platform for developing next-generation bio-based pest management tools in orchards, with potential applications as sprayable mats, patches, or trap inserts [88]. For orchard deployment, these nanofibers can be formulated into liquid suspensions and sprayed directly onto trees, which form thin, durable, and biodegradable mats on leaves and fruits. Such mats can provide controlled release of volatile compounds or other bioactive metabolites produced by yeast, serving diverse functions including attraction of beneficial insects, repellence of pests, or in situ production of biopesticides on the plant surface [88]. Since many insect pests are naturally attracted to yeast-derived volatile organic compounds, immobilizing yeast cells within nanofiber-based trap inserts enables sustained and regulated release of these volatiles over extended periods [77]. This strategy offers an effective and environmentally sustainable approach for pest monitoring and control in orchard ecosystems.

Critical considerations for yeast encapsulation via electrospinning

The viability of live yeast cells relies on their successful encapsulation within nanofibers, which in turn requires optimization of several key parameters to increase encapsulation efficiency and functionality. Figure 4 depicts a decision tree outlining the selection process for suitable solvents and polymers in electrospinning.

Solvent and polymer selection

The solvent choice is the most crucial parameter in electrospinning as there is only a limited range of solvents that can dissolve the chosen polymer and achieve fiber formation [89]. However, the solvents (e.g., chloroform, dichloromethane (DCM), dimethylformamide (DMF), dimethyl sulfoxide (DMSO), tetrahydrofuran (THF), hexafluoroisopropanol (HFIP), formic acid, and acetic acid) that are commonly used to dissolve the polymers for electrospinning pose risk to cellular integrity [90]. Therefore, the use of non-toxic or biocompatible solvents is critical for encapsulating living biological agents [91]. For example, biocompatible decylmonoacyl glycerol (10MAG) and lauryldimethylamino-N-oxide (LDAO) in low-viscosity alkanes have demonstrated effectiveness in preserving the structure and stability of diverse biological macromolecules [92]. Aqueous ethanol (ethanol–water mixtures) is widely regarded as a "green" solvent due to

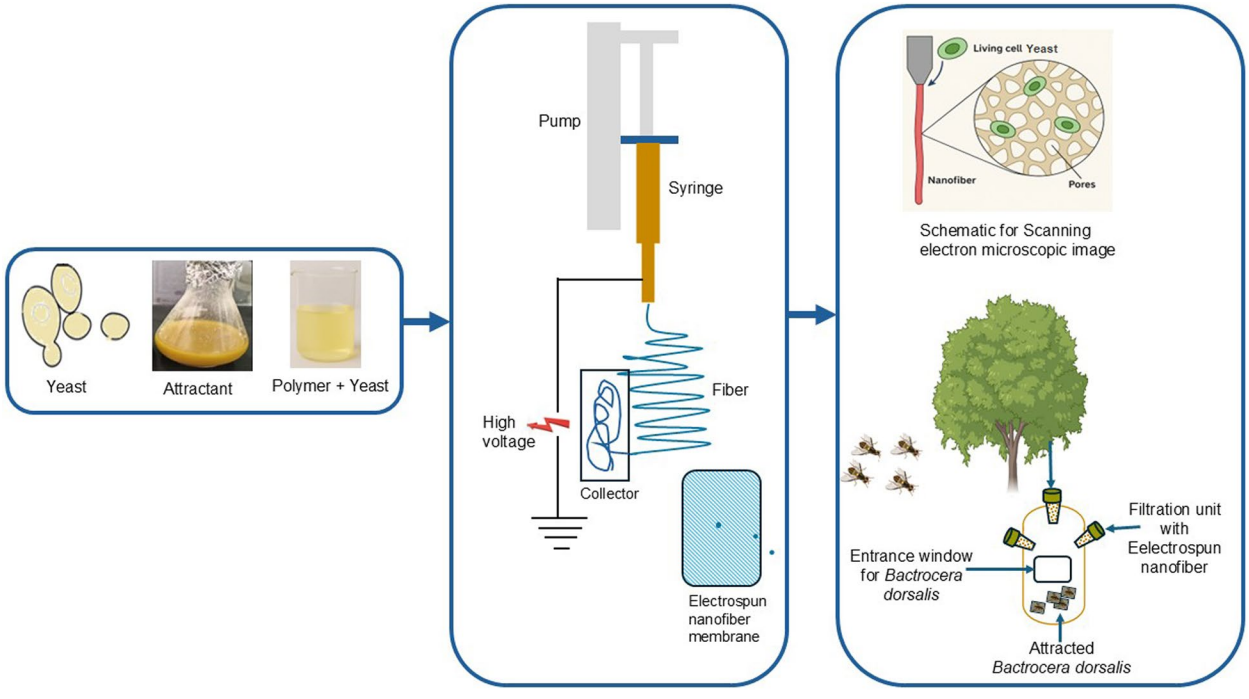


Fig. 3 Schematic representation of the core principle of yeast encapsulation via electrospinning

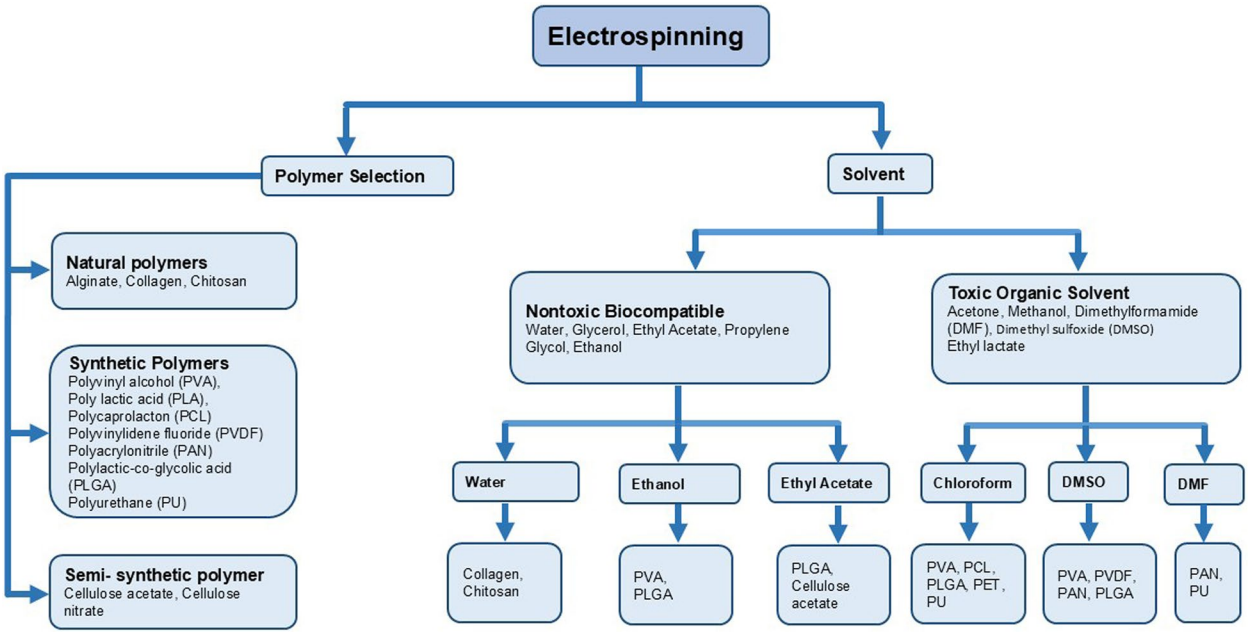


Fig. 4 A decision-tree illustrating solvent and polymer selection for electrospinning

its low toxicity and volatility, which facilitate rapid fiber formation. It is particularly effective for electrospinning zein-based carriers used in the encapsulation of bioactive compounds and probiotics [93, 94]. However, recent study indicates that removing ethanol residues from zein-based systems is challenging, as it can compromise probiotic viability, highlighting a key limitation for cell-compatible applications [95]. In contrast, alginate, frequently blended with polyethylene oxide (PEO) or gelatin, offers an inherently biocompatible alternative. Its rapid ionic gelation provides an immediate protective barrier around cells during and after encapsulation [96, 97]. Nevertheless, pure alginate is challenging to electrospinning due to its high viscosity and polyelectrolytic nature, necessitating the inclusion of carrier polymers such as PEO or gelatin to achieve stable fiber formation [96]

The most common choices of natural polymers include alginate and gelatin while polyvinyl alcohol (PVA) and polylactic acid (PLA) are preferred synthetic polymers [98, 99]. Studies have shown that PVA can effectively protect encapsulated cells, like *Micrococcus luteus* and *Escherichia coli* and maintain their integrity during the electrospinning process [100]. Salalha et al. [101] reported successful encapsulation of bacterial cells (*E. coli* and *Staphylococcus albus*) and bacteriophages (T7, T4, and λ) within electrospun PVA nanofibers. Their findings demonstrated that both bacterial cells and viruses retained viability following electrospinning. Particularly, *S. albus* exhibited 100% viability. Although *E. coli* showed an 81% reduction in viability, pre-treatment with 5% glycerol prior to electrospinning improved *E. coli* viability by up to 50%. In a related study, López-Rubio et al. [102] reported that *Bifidobacterium animalis* Bb12 encapsulated in PVA fibers retained viability for up to 40 days at ambient temperature and 130 days under refrigerated condition. Moreover, the M13 bacteriophage remained structurally intact and infectious post-encapsulation. Liu et al. [103] encapsulated *Pseudomonas fluorescens*, *Zymomonas mobilis*, and *E. coli* in water-insoluble biohybrid nanofibers composed of Pluronic F127 dimethacrylate (FDMA) and polyethylene oxide (PEO). These encapsulated microorganisms remained viable for over two months at – 70 °C. Collectively, these studies showed the significance of optimizing the polymer solution's viscosity, conductivity, and solubility for achieving efficient encapsulation while minimizing cellular damage.

Impact on yeast cell morphology and viability

During the electrospinning yeast cells experience significant physical and chemical stresses that causes change in their original ellipsoidal structure to oblate spheroid structure [77]. These stresses arise from high electric fields, shear forces during solution extrusion, and exposure to solvent [77, 104]. Cumulative stress, especially arising from organic solvents, extreme pH, or dehydration, can significantly reduce cell viability. For instance, Fan et al. [77] demonstrated that immobilized yeast cells remained active for seven days despite some morphological changes [77], yet cell viability was limited to 40%. In lactic acid bacteria, viability has been correlated with hydrophobicity and an elongated morphology [105].

Ongoing research focuses on identifying specific electrospinning conditions that maximize yeast cell stability

and viability within the nanofiber matrix, which is essential for applications requiring sustained release, such as attractants. Efforts to mitigate detrimental effects have shown promise. For example, encapsulation of single yeast cells within nano-organized polyelectrolyte shells, achieved by modifying their adsorption behavior, has demonstrated no detrimental effect on cell metabolic activity or the integrity of the cell and its encapsulating shell [106]. Similarly, encapsulation of brewer's yeast in 0.4% chitosan-coated κ-carrageenan carriers (450 μm in diameter) led to increased α-amino nitrogen consumption compared to free cells. This also resulted in differences in ester and alcohol profiles and altered growth kinetics [107]. However, an increase in carrier diameter due to swelling and the formation of a thicker membrane from the chitosan coating led to reduced cell release. Studies on encapsulated *S. cerevisiae* CBS 8066 revealed decreased growth rate, total RNA/protein contents, biomass, and glycerol yields under anaerobic conditions. In contrast, both ethanol production and glycogen/trehalose content increased [108]. These findings highlight the importance of optimizing electrospinning for biological applications by addressing the complex interplay between encapsulation methods, material properties, and the physiological response of yeast cells.

Environmental and economic sustainability of yeast-based nanofibers

The pervasive economic losses caused by fruit flies globally, stemming from direct crop damage, yield reduction, and trade restrictions, underscore the urgent need for cost-effective and environmentally sustainable pest management solutions. The global losses in major fruit crops caused by fruit flies are estimated to exceed US $3 billion annually [4]. In the United States, invasive fruit fly species pose a serious threat to horticulture, which was valued at approximately US $11.7 billion in 2022 [109]. In Brazil, the Mediterranean fruit fly alone is responsible for an estimated US $242 million in annual economic losses [110]. The problem is equally severe in Africa, where the rapid spread of fruit flies has undermined fruit production and trade; for example, in West Africa, infestations have been reported to cause up to 85% yield losses in mango production [111]. Similarly, in India, economic losses attributed to fruit flies were estimated at to US $305 million in the absence of pest management measures [112].

Traditional reliance on broad-spectrum chemical pesticides often leads to adverse environmental impacts, including biodiversity loss, water and soil contamination, and the development of pesticide resistance in target populations, further compounding economic burdens. Yeast-based attractants delivered via nanofiber encapsulation present a promising paradigm shift toward greater environmental and economic sustainability:

Environmental benefits

Encapsulation of living cells and synthetic polymer can help in reducing chemical footprint by providing highly targeted and localized delivery of attractants, or even biocontrol agents. For example, encapsulated systems release active agents at traps or specific plant sites, thereby reducing the overall volume of pesticides required and minimizing the chemical residues on crops and in the environment [4, 113]. Studies have shown that yeast-based attractants used in combination with low doses of insecticides can achieve control of fruit flies comparable to, or even more effective than, traditional high-dose pesticides [114]. Equally important is the use of biocompatible and biodegradable polymers (e.g., PLA, alginate, cellulose derivatives) as nanofiber carriers themselves degrade naturally over time [115], mitigating concerns about plastic accumulation and microplastic pollution in agricultural ecosystems [116]. Beyond reducing chemical inputs and plastic pollution, yeast-based attractants are also used for species-specific or target a narrow range of pests, reducing harm to non-target beneficial insects (e.g., pollinators, natural enemies) and preserving ecosystem biodiversity. For instance, volatile compounds released by yeasts such as *Hanseniaspora uvarum* and *Saccharomycopsis vini* selectively attract pests like the spotted-wing *Drosophila* (*Drosophila suzukii*) [114], while not stimulating foraging behavior in honeybees, underscoring their safety within integrated pest management frameworks [114].

Extended shelf life and field efficacy

Encapsulation protects the delicate yeast cells and their volatile metabolites from environmental degradation (e.g., UV, heat, moisture), significantly extending the shelf life of the formulation and its effective duration in the field. For instance, a study on the spotted-wing *Drosophila* (*Drosophila suzukii*) found that yeast-based attractants lost their efficacy within a few hours when exposed to direct sunlight [53], whereas encapsulation provides a physical barrier that absorbs the UV light ensures that the yeast remains alive and continues to produce the volatile compounds (e.g., ethanol, acetic acid, and specific esters) that attract the target pests [117]. Encapsulation extends the shelf life of biopesticides and attractants. Without protection, the volatile attractants quickly evaporate, and the yeast cells themselves can degrade [118]. Longer-lasting formulations translate directly into fewer labor hours, and reduced material costs associated with re-application, leading to substantial savings for farmers ultimately reduced application costs. For example, studies on fruit fly control found that using encapsulated bait

allowed farmers to reduce their application frequency from once every 3–5 days to once every 2–4 weeks [4, 115, 119]. More effective and consistent pest control reduces crop damage, leading to higher yields and better-quality produce, which enhances market value and profitability. Effective pest management can help areas maintain pest-free status, reducing the economic burden of quarantine treatments and facilitating smoother international trade. Yeast is an inexpensive and readily available microorganism, and its fermentation processes are generally low-cost. While the carrier polymers (e.g., PVA, biopolymers) add to the cost, the overall formulation is anticipated to be more cost-effective than developing and deploying synthetic, sustained-release pheromones or pesticides.

Ecotoxicological considerations of nanofiber-encapsulated yeast

While offering substantial advantages, the development and deployment of nanofiber-encapsulated yeast for pest management necessitates a thorough assessment of potential ecotoxicological impacts. The high surface area-to-volume ratio of nanofibrous materials, coupled with the biological and metabolic activity of encapsulated yeast, enhances their functionality, biocompatibility, and biodegradability. However, the environmental fate and potential effects of these materials and their biological payload must be carefully considered.

The degradation characteristics of the polymers used in electrospinning (e.g., PVA, PLA, cellulose derivatives) can significantly influence their environmental footprint. Synthetic polymers, even those marketed as biodegradable, may persist in certain environmental compartments and potentially contribute to microplastic pollution if degradation rates are slow or incomplete. In contrast, natural polymers are generally expected to biodegrade more readily and fully assimilate into ecosystems [120]. Comprehensive studies on the degradation pathways and byproducts of specific electrospun polymers in relevant agricultural soil and aquatic environments are crucial.

A key ecotoxicological concern is the controlled release and potential impact of the encapsulated yeast cells or their metabolites on indigenous microbial communities in aquatic or terrestrial environments. While many yeast, species are ubiquitous and generally recognized as safe, the introduction of high concentrations or specific engineered strains could potentially alter local microbial dynamics, nutrient cycling, or even horizontal gene transfer if not managed appropriately. Future research must include environmental fate studies to assess the persistence, dispersion, and degradation kinetics of the nanofiber matrices in different environmental compartments and evaluating the effects of released yeast cells and their metabolites on soil microbiome, aquatic organisms, and non-target insects.

For genetically modified yeasts, stringent risk assessments must be conducted to evaluate potential gene flow and ecological consequences, ensuring their safe deployment. Prioritizing the use of readily biodegradable polymers and conducting thorough pre-release ecological impact assessments are essential to ensure that this innovative pest management strategy is both effective and truly sustainable.

Cost-effectiveness and field feasibility for farmers

For any novel pest management technology to achieve widespread adoption, its cost-effectiveness and practical feasibility for farmers are paramount. While yeast itself is inexpensive, the overall cost of yeast-based nanofiber formulations can be substantial depending on the choice of polymer carriers (e.g., PVA, PLA, other biopolymers) and the manufacturing process. For instance, although biopolymers like PLA are environmentally desirable, the raw material cost can be ~ 2× higher than that of synthetic polymers such polypropylene [121].

Beyond raw materials, the equipment cost, and manufacturing process introduce significant cost barriers for individual farmers or small-scale operations. For example, the current market or industry cost of an industrial-scale electrospinning unit ranges from $200,000 to $500,000 (USD) and high energy consumption of these systems further adds to the operational costs. However, these challenges can be addressed through developing industrial-scale electrospinning processes that can produce nanofibers at a competitive cost per unit and exploring models where centralized facilities produce the nanofiber formulations, which are then distributed to farmers. Further innovation in electrospinning technology to design more energy-efficient and affordable small-scale or portable setups can increase affordability.

Despite the initial investment, the extended efficacy and shelf life provided by nanofiber encapsulation can offset initial higher production costs by significantly reducing the frequency of application, leading to long-term economic savings in labor and materials for farmers. For example, a recent study by Doan et al. [122] demonstrated that a controlled-release pesticide formulation encapsulated in nanofibers required fewer applications over the growing season than conventional sprays, potentially reducing labor and material costs for farmers.

Challenges in implementation of yeast-based nanofibers in pest management

Despite the significant promise of yeast-based nanofibers for fruit fly pest management, several critical challenges must be addressed to translate this technology from concept to widespread agricultural application. These

challenges primarily involve optimizing electrospinning parameters, environmental conditions, solvent selection, and polymer properties.

Electrospinning parameters and process challenges

The morphology and diameter of electrospun fibers are highly sensitive to operational parameters. For instance, applied voltage directly affects fiber diameter, with higher voltages producing thicker fibers often accompanied by bead defects [123]. Similarly, flow rate influences fiber formation: rates between 2 and 4.5 mL/h increase fiber diameter, while rates exceeding 4.5 mL/h reduce it [124]. The distance between the needle and collector also plays a critical role; increasing this distance generally produces thinner, more uniform fibers with broader deposition areas, whereas excessively large or small distances lead to defects or oversized fibers [125, 126]. Effective electrospinning is typically achieved at a needle-to-collector distance of 10–20 cm, however, the specific optimal distance can vary depending on other factors such as the polymer solution, applied voltage, and desired fiber morphology [16, 127].

Environmental sensitivity

Electrospinning is highly sensitive to environmental conditions such as temperature, humidity, and air pressure. Temperature influences solvent evaporation, solution viscosity, and conductivity, all of which determine fiber diameter and porosity [128]. Elevated temperatures accelerate solvent evaporation, reduce solution viscosity, and increase conductivity, collectively promoting formation of porous fibers [129]. Humidity inversely affects temperature; reduced humidity accelerates solvent evaporation, modifying pore size and fiber porosity [130]. Air pressure, typically atmospheric during electrospinning, also impacts fiber morphology by altering evaporation rates and deposition patterns. Fluctuations in air pressure can compromise fiber consistency and uniformity [131].

Solvent selection

The choice of solvents is critical, as traditional solvents such as chloroform, trifluoroethanol, and dimethylformamide are toxic to living cells [132] and increasingly restricted under regulations such as Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) in the European Union. Green solvents—including acetic acid, ethanol, water, and their mixtures—provide safer alternatives suitable for incorporating viable cells into electrospun fibers [93].

Polymer selection

Polymer properties significantly influence fiber formation. Selection depends on solubility in the chosen solvent, polymer concentration, and molecular weight [133]. Common polymers used in electrospinning include natural polymers (e.g., alginate, collagen), semi-synthetic polymers (e.g., cellulose acetate), and synthetic polymers (e.g., polyvinyl alcohol, polylactic acid, polycaprolactone). Low polymer concentrations can disrupt continuous fiber formation, whereas high concentrations increase solution viscosity, affecting flow and fiber diameter [129, 134]. The conductivity of the polymer solution also affects fiber morphology; higher conductivity generally reduces fiber diameter due to greater tensile forces in the electric field [126]. Conductivity can be modulated by adding salts such as lithium chloride, ammonium chloride, sodium phosphate, or potassium phosphate [135].

Future research directions

The development of effective and sustainable strategies for fruit fly management requires innovative approaches that transcend conventional control methods. One promising approach involves combining nanofiber technology with other advanced materials, such as 3D-printed nanofiber constructs, to create more complex architectures for ultra-controlled release or multi-functional pest management devices. Integration of 'attract-and-kill' to develop nanofibers that co-encapsulate yeast attractants with highly targeted, eco-friendly insecticides or entomopathogenic organisms (e.g., fungi, bacteria, nematodes) to achieve both attraction and mortality. Furthermore, investigation should focus on the potential for broad-spectrum yeast-based attractants or formulations targeting multiple pest species simultaneously. Additionally, researchers can integrate nanofiber systems with miniaturized sensors to monitor environmental conditions or attractant release dynamics in real-time, enabling smart pest management. Researchers can also use advanced metabolomics to precisely characterize the volatile profiles of encapsulated yeast strains under different conditions ensuring maximum and sustained attractiveness.

While electrospinning is a versatile and cost-effective method for producing nanofibers, its high sensitivity to parameters such as polymer concentration, viscosity, and conductivity limits its commercial use [19]. The integration of machine learning (ML) and artificial intelligence (AI) provide a powerful solution for optimizing the electrospinning process thereby eliminating the reliance on time-consuming trial-and-error experimentation [136, 137]. For example, Luque et al. [138] used an Artificial Neural Network (ANN) model and trained on experimental data to predict optimal parameters for nanofibers. The model achieved a low Root Mean Square Error (RMSE) of 7.77% thereby significantly enhancing process efficiency. Furthermore, using Shapley Additive Explanations (SHAP) analysis, Sukpancharoen et al. [139] were able to statistically quantify the influence of polymer concentration, applied voltage, and feed rate on nanofiber diameter.

Complementing AI, the Internet of Things (IoT) provides the critical real-time data of these optimized conditions [140]. IoT sensors can be integrated directly into the electrospinning setup to monitor and regulate environmental and process parameters. For instance, the development of an IoT-based Electrospinning Temperature Control (ETC) system. This system utilizes a DHT21 sensor to monitor environmental temperature with high accuracy (94.95%) and precision (98.93%) [140]. A microcontroller, such as an ESP32, processes this data and transmits it to a cloud platform, enabling remote monitoring and control via a smartphone application.

The synergy between AI and IoT creates a powerful, closed-loop feedback system. The AI model predicts optimal conditions, and IoT sensors provide real-time data to monitor any deviations from the optimized parameters [138]. This real-time data stream can not only be used to correct the process as it occurs but also continuously retrains and refines the AI model. This constant adaptation allows the system to compensate for unforeseen variables and environmental fluctuations, transforming the electrospinning process from a sensitive, manual operation into a robust, self-optimizing manufacturing line.

Moving forward, it is essential to conduct Life-Cycle Assessment (LCA) studies to comprehensively evaluate the environmental impacts of traditional techniques in fruit fly pest management. Previous LCA studies have already shown that precision agriculture strategies, including the Male Annihilation Technique (MAT) and the Sterile Insect Technique (SIT), have significantly lower environmental footprints compared to conventional agricultural practices [141, 142]. Integrating such approaches within a precision agriculture framework will be key to developing sustainable pest management systems [143, 144]. By addressing these challenges through interdisciplinary research will be crucial for the successful translation of yeast-based nanofiber technology into a truly sustainable and effective tool for fruit fly pest management globally.

Conclusions

The combination of yeast-based attractants and electrospinning technology offers a paradigm shift in sustainable fruit fly pest management. This multidisciplinary approach harnesses the biological effectiveness of yeast cells and capabilities of nanofiber encapsulation to address the current insect pest management technology. Yeast-based electrospun nanofibers hold immense potential in enhancing pest control by providing sustainable release bioactive compounds, offering prolonged efficacy and improved pest control. These natural and eco-friendly electrospun nanofibers improve sustainability through reduced dependency on chemical pesticides, minimized environmental pollution, and utilizing biodegradable components. While the conceptual framework is convincing and results of previous studies have laid the groundwork for potential impact in insect pest management, its widespread use requires addressing the challenges related to large-scale production, controlled release in varied environmental conditions, ecotoxicological assessment, and economic viability. Furthermore, to address the existing gaps in development of attractant, optimal electrospun nanofibers and controlled delivery system, future research should focus on interdisciplinary collaborations. Finally, the idea of using yeast-based nanofibers developed via electrospinning has the potential to transform fruit fly control while promoting environmental conservation and strengthening the economic sustainability of global fruit production.

Abbreviations

2-PEA	2-Phenethyl Alcohol
AI	Artificial Intelligence
ANN	Artificial Neural Network
Bt	Bacillus thuringiensis
CA	Cellulose acetate
CpGV	Cydia Pomonella Granulovirus
DCM	Dichloromethane
DEET	N, N-diethyl-m-toluamide
DEF	N, N-diethylformamide
DMF	Dimethylformamide
DM-PEO	Dextran modified poly(ethylene oxide)
DMSO	Dimethyl Sulfoxide
dsRNA	Double-Stranded RNA
ETC	Electrospinning Temperature Control
FDMA	F127 Dimethacrylate
Gm-DES	Desaturase gene from the moth Grapholita molesta
Gm-FAR	Fatty Acyl-CoA Reductase (FAR) gene from Grapholita molesta
HFIP	Hexafluoroisopropanol
IoT	Internet of Things
LCA	Life-Cycle Assessment
LDAO	Lauryldimethylamino-N-oxide
MAG	Decylmonoacyl glycerol
MAT	Male Annihilation Technique
ML	Machine Learning
MOFs	Metal-Organic Frameworks
PAAm	Polyacrylamide
PCL	Poly(caprolactone)
PEA	Pea protein
PEO	Polyethylene Oxide
PET	Polyethylene terephthalate
PLA	Polylactic Acid
PLGA	Poly(lactic-co-glycolic acid)
PLLA	Poly(L-lactic acid)
PVA	Polyvinyl Alcohol
REACH	Registration, Evaluation, Authorisation and Restriction of Chemicals
RMSE	Root Mean Square Error
RNAi	RNA Interference
SHAP	Shapley Additive Explanations
SIT	Sterile Insect Technique
THF	Tetrahydrofuran
VOCs	Volatile Organic Compounds

Authors' contributions

SR conceived the manuscript and prepared the initial draft. GK revised the content and created the figure. RK helped significantly in addressing the reviewer comments. All authors reviewed and approved the final version of the manuscript.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Competing interests
The authors declare no competing interests.

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